

Adoption playbook — putting SSSF to work on real code

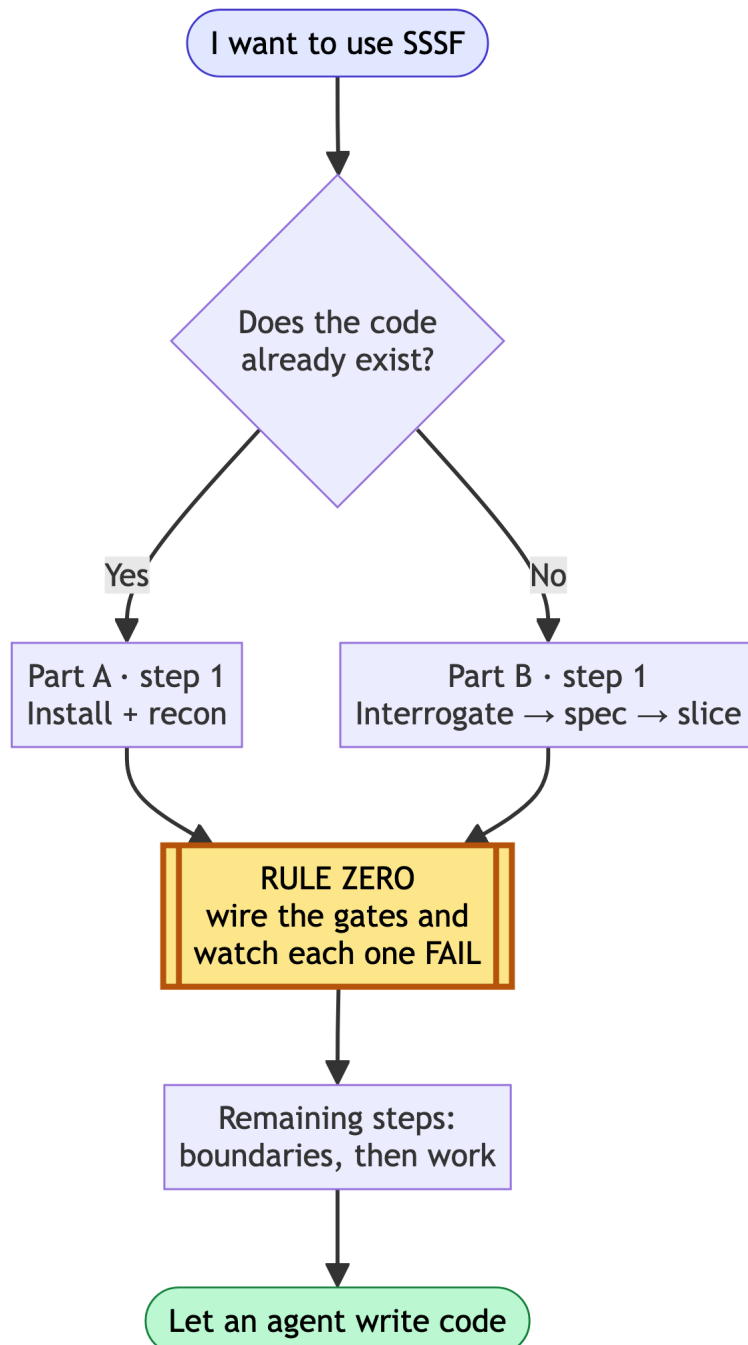
Adoption playbook

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Adoption playbook

Two situations: a codebase that already exists, and a project that does not yet. Both start from the same non-negotiable rule.



Rule zero sits early in both parts, not at the end: gates are wired before any agent writes a line.

Rule zero: wire the gates before an agent writes a line

A freshly stamped factory ships every quality block as an `echo` that exits 0. Run a workflow against it and you get a green trace that proves nothing — the agent's claim went unchecked and the run *reported success*. This is worse than having no factory, because it manufactures confidence.

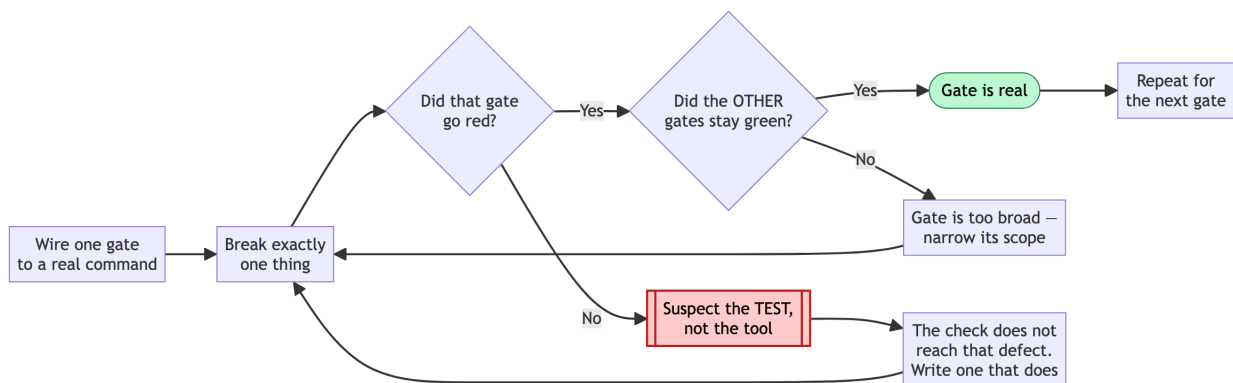
So the first thing you do in any repo, before any agent writes anything:

```
just quality "baseline"           # zero agents – look at what it claims
```

If it says `PLACEHOLDER`, the gates are fake. Wire them in `adws/adw_modules/quality.py`, then **prove each one can fail**:

```
# break exactly one thing, run the gates, confirm exactly one goes red,
  restore
```

A gate you have not watched fail is not a gate. If a gate stays green against a deliberately broken build, treat the *check* as the defect, not the tool: the check does not reach that class of breakage. Do this step before anything else in the adoption — every later step assumes the gates are real.



Rules for the commands themselves: `argv` is a **list**, never a shell string; call binaries by bare name so they resolve in the operator's environment; and put scope and strictness in your config files (`pyproject.toml`, `package.json`) rather than in the `argv`, so the gate and a local run can never disagree.

Part A — an existing codebase

Order: install + recon → wire the gates (rule zero) → set boundaries → interrogate → work.

A1. Recon before spending

```
cd /path/to/your/repo
uv run /path/to/ssf/.claude/skills/ssf/scripts/install.py
git add -A && git commit -m "Install SSF"      # keep the tree clean, see A3
just scout "map this codebase: entry points, layers, test setup, build
          commands"
```

`just scout` is read-only. Its findings land in `adws/adw_data/sessions/<adw_id>/context_handoff/scout_findings.md`. Read them. You are checking whether the agent understood your repo before you let it change anything.

A2. Wire the gates (rule zero)

Point `test`, `lint` and `typecheck` at your real commands. Delete blocks you do not have — a `build` block running `echo` is a phantom check, and `run_quality()` reports it as passed. (That behaviour is in `adws/adw_modules/quality.py` — read `run_quality()` if you want to see it yourself rather than take this on trust.)

If your suite is slow, scope the test gate to a fast subset now and widen later. A gate nobody waits for gets disabled, and a disabled gate is a placeholder with extra steps.

A3. Set the boundaries

Two different mechanisms, and people conflate them:

- `tools` is a capability list — what the agent *can* do.
- `writes` is a boundary — what it *may change in the repo*, enforced after every call by diffing the tree.

Both are documented behaviour of the agent runner in `adws/adw_modules/` — grep for `writes` there to see the enforcement point.

For an existing codebase, tighten `writes` before your first build run. Give the documenter `docs/` and `**/*.md`. Give the planner `specs/`. Leave the builder unrestricted only if you are ready to review its whole diff.

`protected_files` is off-limits to everyone: keep `adws/adw_modules/`, `adws/adw_sssf_config/` and `adws/adw_*.py` in it, so no agent can edit the machinery that grades its own work. Add your CI config and any secrets path.

Sharp edge: the commit phase runs `git add -A` — it stages the *entire* working tree. Grep for `add -A` in `adws/adw_modules/` to confirm on your version. Start every run from a clean tree, or your unrelated work in progress lands in the agent's commit.

A4. Interrogate before you plan

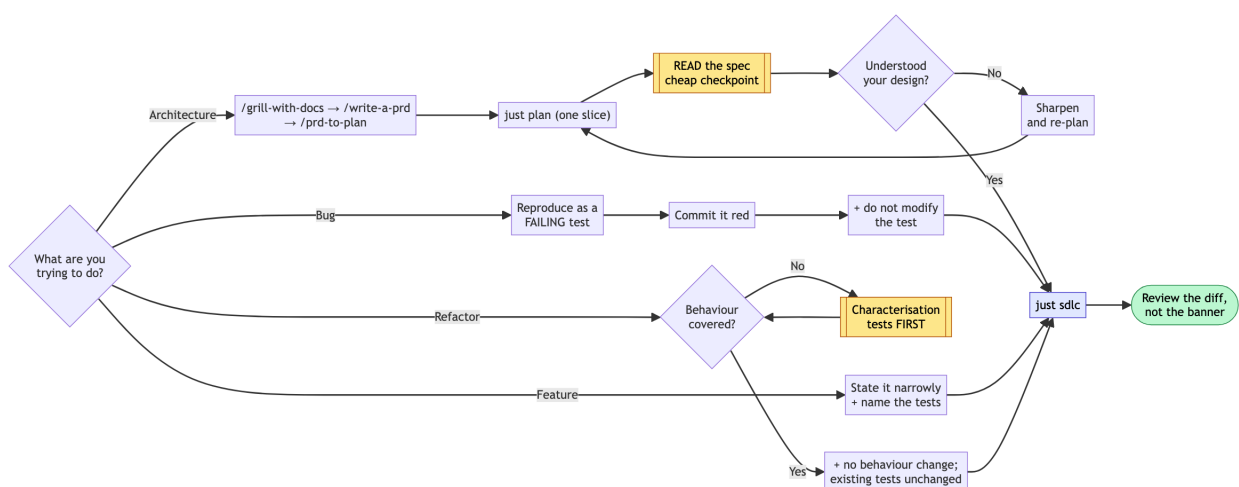
Same order as Part B, applied to code that already exists: interrogation comes before spec-writing, spec-writing before slicing.

Do this interactively in Claude Code, where your installed skills are available:

```
/grill-with-docs  interrogate your own intent against the scout findings from
                  A1 – what actually breaks today, what must not change, what
                  you are assuming about the existing design. Writes the
                  resolved vocabulary to CONTEXT.md/docs/adr/ as you go.
/write-a-prd      synthesise that conversation into a spec.
/prd-to-plan      break the spec into slices, each independently shippable.
```

Then hand one slice to SSSF. `just plan` is SSSF's own synthesis step, and it is downstream of your thinking, not a replacement for it.

A5. The four jobs



Improve architecture. Do not hand this to a build workflow. Grill, spec and slice first (A4), then plan one slice, read the plan, then decide:

```
just plan "extract the payment provider calls behind one interface; no
          behaviour change"
# read specs/<adw_id>*.md yourself – this is the cheap checkpoint
just sdhc "<same slice, once the plan is right>"
```

`just plan` runs the planner alone. Finding out the agent misunderstood your architecture costs one planner run instead of a full build plus a bad diff.

Find and fix bugs. The fix loop is a repair machine, so give it something red to repair. Reproduce the bug as a failing test first — by hand, or with a cheap agent — commit it red, then:

```
just sdhc "make the failing test in tests/test_x.py pass; do not modify the
          test"
```

The `do not modify the test` clause matters: the builder has write access, and deleting an assertion is the cheapest path to green. Constrain it explicitly rather than relying on virtue you have not asked for.

Implement a feature. The main line. Keep the slice narrow enough that one plan can express it:

```
just sdhc "add X; follow existing conventions; add tests covering A, B and C"
```

Cost scales with plan size. Measure your own first few runs with `just sessions` before you budget the rest.

Refactor into reusable abstractions. The most dangerous job, because “working” and “unchanged” are different claims. Refactoring is only safe behind tests that pin *current* behaviour, so:

1. Check coverage of the code you are about to move. If it is thin, spend a run adding characterisation tests **first**, and commit them.
2. Then refactor with an explicit no-behaviour-change constraint:

```
just sdhc "extract the discount rules in the cart-pricing module behind one
          interface. No behaviour change: every existing test must pass
          unchanged. Do not modify or delete any existing test."
```

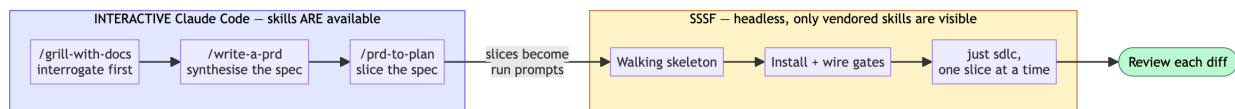
The green suite is your evidence the refactor was safe. Without it you are not refactoring, you are rewriting and hoping.

A6. Read what happened

```
just sessions          # last runs, cost, status
just phases <adw_id>   # which phase did what
```

Check cost per run early and often, and build your own baseline from those numbers — no figure from someone else's repo transfers to yours. If one agent is eating most of your spend, try a cheaper model on it and compare both the price *and* the output it produces. Do not assume the expensive model plans better; run the same prompt through both on your own repo and read the two specs.

Part B — a new project



B1. Think before there is code (interactive, not SSSF)

SSSF is an execution engine. It is the wrong tool for deciding what to build.

Do this part **interactively** in Claude Code, where your installed skills are available. The order matters, and it is the reverse of what most people assume:

```
/grill-with-docs  interrogate relentlessly, round after round, until nothing
                  important is left silently assumed. This happens BEFORE any
                  spec exists – it is how you reach shared understanding.
                  Writes the resolved vocabulary to CONTEXT.md/docs/adr/ as
                  you go, so it survives past this conversation.

/write-a-prd      synthesise that conversation into a structured spec.
                  No interrogation here; it already happened.

/prd-to-plan      break the spec into slices, each independently shippable and
                  verifiable, with explicit blocking order.
```


Skipping the grill and starting at the spec produces a document that reads well and encodes assumptions nobody tested. The spec step is downstream of the thinking, not a substitute for it.

What the difference looks like. Same feature request — “let users export their data” — written two ways.

Straight to spec, no interrogation:

Add a data export feature. Users can export their data as CSV.
Acceptance: user clicks Export, receives a CSV file.

After a grill that asked which data, how much, who is allowed, and what happens when it is slow:

Export scope: only records the requesting user owns (confirmed: no team-shared records in v1).
Volume: 95th percentile account is ~40k rows → synchronous download times out.
Export is a background job; user gets an email link. Link expires in 24h.
Format: CSV only. JSON was requested and deferred — decided in the grill, not discovered mid-build.
Excluded: soft-deleted records, other users' PII in shared audit rows.
Acceptance: 40k-row account exports without timeout; a user cannot export a record they do not own; expired link returns 410.

The second is not longer for its own sake. Every extra line is a question that would otherwise have been answered by the agent guessing, mid-build.

Why still interactive: SSSF agents run with `--setting-sources ''`, so they cannot see the skills installed on your machine. Grep the agent launch args in `adws/adw_modules/` to confirm. That has always been true and remains true. The addition is the `skill_engineering`: config key: a *specific named* skill can be vendored into the repo and attached to one agent (see “Where the two layers sit”), so the two layers can meet — but only for skills you choose on purpose, and only for `claude_code` agents. Your general interactive toolkit is still yours alone.

B2. Make it a repo with a walking skeleton

```
mkdir project && cd project && git init
```

Before installing SSSF, build the thinnest thing that runs and is tested — by hand or in one interactive session:

- one real module with one real function
- a test runner that works, with at least one passing test
- lint and typecheck configured

This is the target the factory needs. An empty repo gives the gates nothing to check, and you are back to placeholder-green.

```
git add -A && git commit -m "Walking skeleton"
```

B3. Install and wire (rule zero)

```
uv run /path/to/ssf/.claude/skills/ssf/scripts/install.py
# wire quality.py to your real commands, then prove each gate fails
just quality "gates wired"
git add -A && git commit -m "Install SSSF and wire the gates"
```

B4. Tune the roster

Start with everything on `claude_code` — no API key, uses your logged-in session. Then:

- `writes: []` on every read-only agent (scout, reviewer)
- documenter limited to docs paths
- planner limited to `specs/`
- cheap model on scout and documenter, stronger on planner and reviewer
- add your CI config to `protected_files`

B5. Run the slices, one at a time

Feed the slices from B1 in one at a time:

```
just sdlc "<one slice, stated as an outcome with its acceptance criteria>"
```

Review every diff before starting the next. The factory removes typing, not judgement. One slice per run keeps the blast radius small and the plan cheap.

The standing checklist

Before any run that writes:

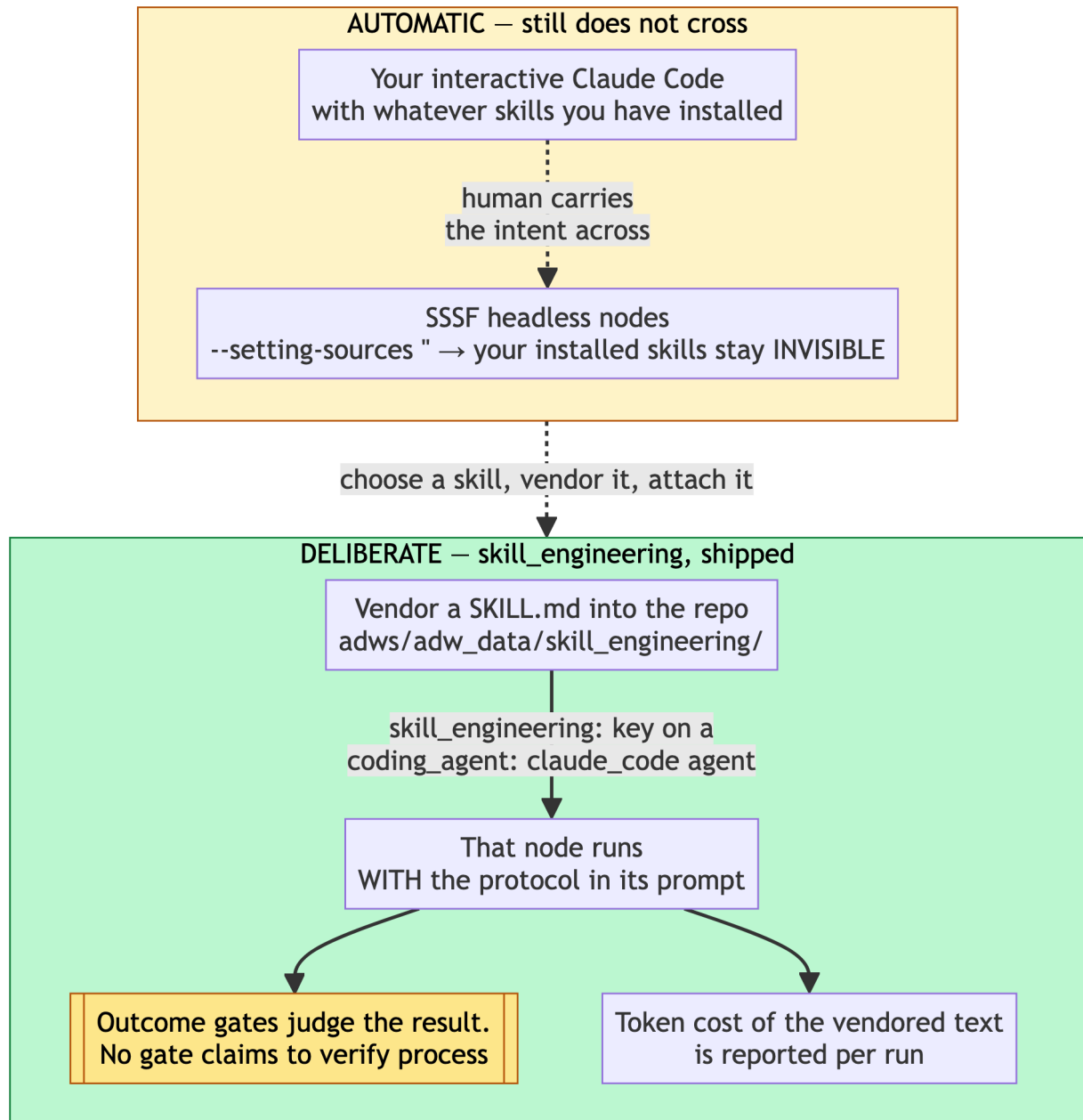
- ☐ tree is clean (`git status`) — the commit phase stages everything
- ☐ gates wired, and each one watched failing at least once
- ☐ `writes` set for every agent that should not touch the whole repo
- ☐ `protected_files` covers CI config and the factory's own machinery
- ☐ you are on a branch you are willing to throw away

After:

- ☐ read the diff, not just the green banner
- ☐ `just sessions` — did it cost what you expected
- ☐ if a gate never went red across several runs, suspect the gate

Where the two layers sit

Skill vendoring ships in the current version of SSSF: the vendoring command and the `skill_engineering:` config key are both present. The layers meet on purpose, one skill at a time:



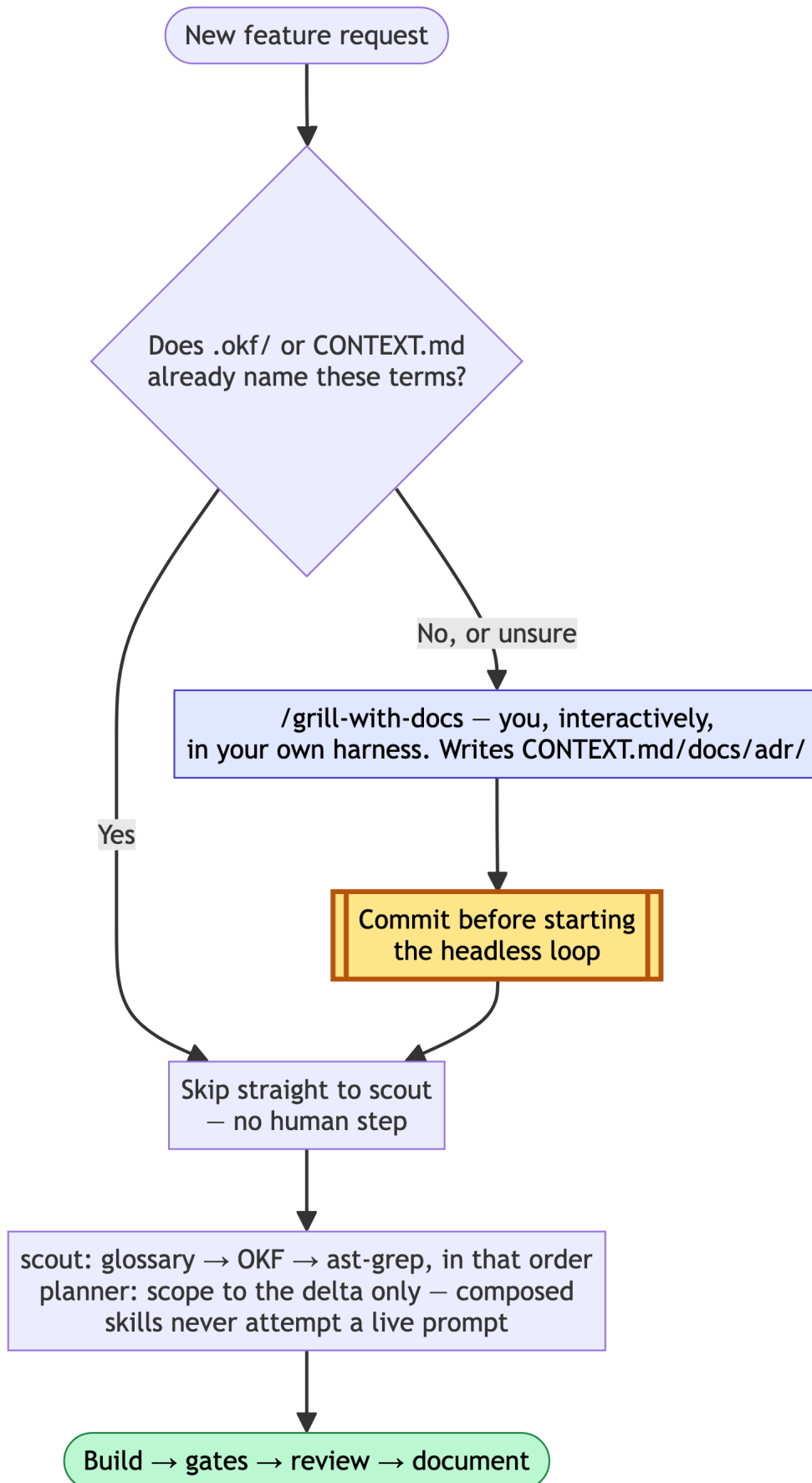
How it works. Each claim below is documented behaviour you can verify in `adws/adw_sssf_config/ssf.config.yaml` and the agent runner in `adws/adw_modules/`:

- **Vendor** the skill file into the repo with the vendoring command. The text now lives in your tree, versioned with your code.
- **Attach** it with the `skill_engineering:` config key on an agent in `ssf.config.yaml`. That agent's prompt carries the protocol.
- **claude_code only.** Agents configured as `pi` or `agy` ignore the key — check where `skill_engineering` is read in `adws/adw_modules/` to confirm on your version.

- **Cost is visible.** Vendored skill text is real tokens on every call that agent makes, and the run reports it in the session record `just sessions` reads. Attach protocols you want, not every protocol you own.

Part C — going dark: bootstrap vocabulary once, then let the loop run headless

Parts A and B get you to `just sdlc` running safely. This part is what “dark factory” adds on top: minimal human intervention *across many features*, not just within one run. Two problems block that, and both have the same shape — something a human normally supplies mid-conversation has to be supplied *before* the headless loop starts instead, or the loop stalls or guesses.



Problem 1 — the interview `write-a-prd` wants doesn't have anyone to answer it

`write-a-prd` (and `wayfinder`'s "ask the user how to proceed" fallback) are written for an interactive session — a human on the other end who answers back. Vendored under `skill_engineering:` and run by a headless `claude_code` node, there is no one there. Best case the model role-plays both sides and you get a low-fidelity PRD with none of the interview's real value; worst case it tries to prompt and the run hangs waiting for input that never comes.

Do not delete the interview — relocate it. The interview is genuinely how a project converges on shared vocabulary (what does "cancellation" mean here, is "account" the Customer or the User), and that can't be skipped just because a run is unattended. Split it into two phases that run in different modes:

- **Bootstrap, human-supervised, outside the headless loop.** When a feature introduces domain concepts your project hasn't named yet, run `/grill-with-docs` yourself — interactively, in whichever harness you're in. It composes `grilling` (the relentless interview) with `domain-modeling` (writes the resolved vocabulary into `CONTEXT.md / docs/adr/` — durable and citable, not prose trapped in one PRD). Approve it, then **commit** `CONTEXT.md / docs/adr/` **before starting** `just sdlc` — scout and planner read them from the working tree, not from any live session. Skipping the commit makes the whole bootstrap step invisible to the pipeline.
- **AFK, headless, the common case.** If the request only composes concepts `CONTEXT.md` already names, skip bootstrap entirely — there's nothing left to interview about. `write-a-prd` runs unmodified (do not edit the vendored copy) and degrades to pure PRD-formatting from already-agreed vocabulary. Add this to your `skill_engineering`-attached agent's own `system.md` (not the vendored skill file) so it knows what to do when the composed skill text below it says "ask the user":
 - `wayfinder` finds no fog → continue directly into `write-a-prd`, don't stop and ask how to proceed.
 - `write-a-prd` never prompts. If it hits a genuine ambiguity `CONTEXT.md` can't resolve, it says so in the plan and in `notes_for_next_agent`, proposes a best-guess term explicitly labeled provisional, and flags that a human should run the bootstrap interview before the plan is final — it does not guess silently and does not attempt to prompt.

- `prd-to-plan` 's “ask the user to paste it” branch never applies in-loop — the PRD is always already in context from the prior step in the same composed prompt.

`grill-with-docs` should stay **out of** `skill_engineering/` — its `disable-model-invocation: true` is load-bearing, not an oversight. Vending it risks it landing in an agent's composed prompt and firing headless despite that flag, which is exactly the failure this split exists to prevent.

Problem 2 — nothing stops a plan from re-implementing what already exists

Neither `prompt_engineering/` nor any shipped `skill_engineering/` skill tells scout or the planner to check what's already built before scoping new work. Left unfixed, every feature costs a full plan-and-build cycle even when 80% of it already exists — the opposite of minimal diff.

Add two lookups to **scout's** `system.md` , before any other search:

1. **Glossary first**, if `CONTEXT.md / docs/adr/` exist (see Problem 1) — note any request term missing or conflicting; that's a signal for the planner, not something scout resolves itself.
2. **A structured knowledge source second**, if your project maintains one — an OKF-style knowledge bundle, a `CONTEXT.md` -linked concept map, or similar. Report what's *already implemented* and how, cited by concept path, not line numbers, which drift.
3. **A structural code search third** (e.g. `ast-grep`) to confirm the knowledge source's claims against actual code. Documentation can lag; ground truth doesn't. When they disagree, trust the code and flag the doc as stale.

Then in **planner's** `system.md` : read scout's findings first, and scope the plan to the delta only — anything scout already marked implemented is explicitly out of scope, not silently re-touched.

None of this needs a new SSSF agent role or a new `skill_engineering` entry — `ast-grep` and a knowledge-source lookup are tool-usage patterns baked into scout's own instructions, the same way `writes` and `protected_files` already are. They don't belong in the same composition list as `wayfinder / write-a-prd / prd-to-plan / tdd` , which are planning methodologies.

Definition of done, extended

Everything in the standing checklist above, plus:

☐

If bootstrap ran, `CONTEXT.md / docs/adr/` are committed and predate the `just sdlc` run that used them.

☐

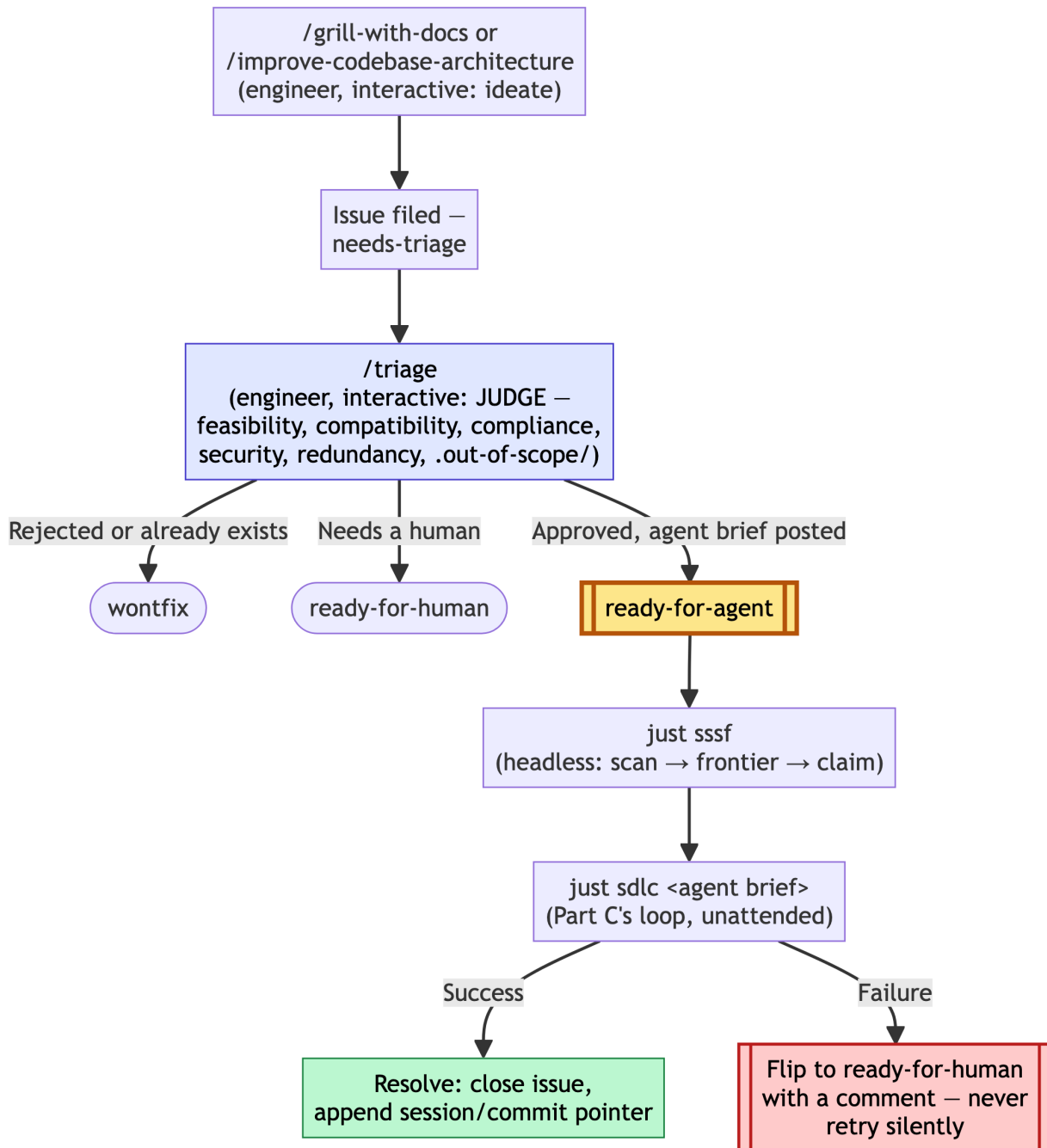
The plan explicitly excludes anything scout marked as already implemented — check `specs/<adw_id>_*.md` names what it's *not* doing, not just what it is.

☐

Your knowledge source (OKF, concept map, whatever you use) got updated by the documenter stage if the feature changed anything it describes — otherwise the next feature's discovery step is reading stale claims.

Part D — the queue: from `ready-for-agent` to shipped

Part C gets one feature through the headless loop unattended. This part is what turns that into a standing queue: an engineer files or triages work whenever they want, and something — `just sssf`, watching — picks it up and ships it without anyone manually running `just sdlc` per item.



The judgment call stays interactive, on purpose

It's tempting to have `just sssf` itself analyze a raw issue's feasibility, compatibility with what's already live, and compliance/security against `CLAUDE.md` / `AGENTS.md` , `CONTEXT.md` , your knowledge source, and `docs/` /plan files — then decide whether to build it. Don't: that recreates the exact failure Part C already fixed once. `write-a-prd` 's interview had no one to answer it headless; an unattended loop making its own security/compliance judgment calls has no one to catch it when it's wrong, either.

The `triage` skill (if installed) already does this analysis — feasibility, redundancy against existing implementation, `.out-of-scope/` prior-rejection checks, compliance — but it is explicitly interactive: it waits for maintainer direction at every step and never runs unattended. Its terminal state for approved work, `ready-for-agent`, means a durable **agent brief** has been posted to the issue. That’s the same pattern as `grill-with-docs`: the judgment happens once, by a human, and only the *result* — not the judgment process — becomes something a headless loop can safely consume later.

So: `just sssf` **only ever picks up issues already in `ready-for-agent` state**. Everything that decides whether something is safe, compatible, and compliant to build happens upstream of that label, interactively. Scout and planner’s Part C wiring (glossary/OKF/ast-grep discovery, delta-scoping) still runs inside every `just sdlc` call underneath this — defense-in-depth, not a substitute for the triage gate.

The queue mechanism already exists — reuse it

Don’t build a new work-item format. `docs/agents/issue-tracker.md`’s “Wayfinding operations” section (written by `/setup-matt-pocock-skills`, see Part A/B) already defines everything a queue needs: a **map** + numbered **child** files, a `Blocked by:` line, a **frontier** rule (“scan for files that are open, unblocked, and unclaimed; first by number wins”), and **claim/resolve** semantics. That’s dependency-ordered work-item tracking, full stop — built for a human working research tickets, but the mechanism doesn’t care who’s claiming. `just sssf` reuses it verbatim: same frontier scan, same claim-before-work, same resolve-after-work, just performed by an agent instead of a human.

This also closes a real gap: a `prd-to-plan` plan’s individual **phases** (Part A5/B1) have no atomic, independently-pickable file of their own by default — only the feature’s `spec.md` does. For a queue to claim work safely, each phase needs its own issue file, with a `Blocked by:` line expressing phase ordering (Phase 2 blocked by Phase 1) the same way a wayfinder ticket blocks on another.

What `just sssf` does, concretely

A polling loop, tracker-aware (reads `docs/agents/issue-tracker.md` to know whether issues live on GitHub or under `.scratch/`):

1. **Scan** for `ready-for-agent` issues — `gh issue list --label ready-for-agent` or `grep Status: ready-for-agent` across `.scratch/*/issues/*.md`.

2. **Frontier:** filter to unblocked (every `Blocked by:` target already resolved) and unclaimed; oldest/lowest-numbered first.
3. **Claim:** set `Status: claimed` before touching anything else — same as wayfinder, so two concurrent `just sssf` runs never double-pick.
4. **Dispatch:** extract the agent brief, run `just sdlc "<brief>"`.
5. **Resolve:** on success, close the issue and append a pointer (session id, commit, cost) — mirroring wayfinder's "append a context pointer to the map." On failure, flip to `ready-for-human` with a comment explaining what broke. Never retry silently — a loop that keeps re-attempting the same failure burns money without anyone finding out until later.
6. **Loop** on an interval, or — more composable — run once-and-exit, invoked periodically by `cron / launchd / CI` outside SSSF's own scope, rather than the factory growing its own daemon.

Definition of done, extended again

Everything in Part C's definition of done, plus:



Every issue `just sssf` picked up was in `ready-for-agent` state when it claimed it — never a raw or `needs-triage` issue.



A failed run is `ready-for-human` with a comment explaining what broke, not silently retried or left `claimed` forever.



Each `prd-to-plan` phase intended for the queue has its own issue file with `Blocked by:` expressing phase order, not just a shared `plan.md`.

What this playbook does not claim

- That agent output needs no review. Agents fabricate confidently, including about their own work. Read the diff before you keep it.
- That vendoring a skill makes an agent follow it. Attaching `tdd` puts the protocol in the prompt. Nothing verifies the agent worked test-first — the gates judge the outcome, not the process. That distinction is the point.
- That any cost figure transfers to your repo. Language, suite size, and repo shape dominate. Measure your own on a throwaway branch before budgeting.

- That `just sssf` (Part D) exists as shipped code. It's a design, reasoned through against the tracker mechanisms this playbook already documents — build it before pointing a scheduler at a repo expecting it.

Version history

Version	Date	Changes
4.0	2026-09-09	Added Part D — the queue: turns Part C's single unattended run into a standing queue. Keeps feasibility/compatibility/compliance/security judgment interactive via the <code>triage</code> skill (terminal state <code>ready-for-agent</code> posts a durable agent brief, same bootstrap-then-headless pattern as <code>grill-with-docs</code>); <code>just sssf</code> only ever claims <code>already-ready-for-agent</code> work. Reuses wayfinder's existing map/child/frontier/claim/resolve mechanism as the queue rather than inventing a new one, and flags the gap it exposes: <code>prd-to-plan</code> phases need their own issue files with <code>Blocked by:</code> to be queue-pickable. Design only, not yet built.

Version	Date	Changes
3.1	2026-09-08	Replaced every <code>/grill-me</code> reference (A4, A5, B1) with <code>/grill-with-docs</code>. <code>grill-me</code> is a bare alias for the <code>grilling</code> interview with no artifact output; <code>grill-with-docs</code> composes the same interview with <code>domain-modeling</code>, writing resolved vocabulary to <code>CONTEXT.md / docs/adr/</code>. Part C's AFK mechanism depends on that vocabulary existing on disk for scout/planner to read — <code>grill-me</code> alone can't produce it, so the playbook now names one interview skill throughout, and it's the AFK-sufficient one.

Version	Date	Changes
3.0	2026-09-08	Added Part C — going dark: relocates <code>write-a-prd / wayfinder</code> 's interactive interview to a human-supervised bootstrap phase (<code>/grill-with-docs</code> , writes <code>CONTEXT.md / docs/adr/</code> , committed before the headless loop starts) so the AFK loop never needs to prompt live; adds glossary → knowledge-source → structural-search discovery to scout before planning, so plans scope to the delta instead of re-implementing what exists. Generalized from a same-session design pass on a real project (<code>weather-report</code> 's <code>docs/agents/dark-factory-protocol.md</code>), which stays as that project's concrete instance of this part.

Version	Date	Changes
2.1	2026-08-31	Removed the last cost figures so the measure-it-yourself stance is consistent. Reframed gate predictions as conditional instructions. Propagated the grill → spec → slice order into Part A as a new A4 step. Unified naming on <code>/grill-me</code>, <code>/write-a-prd</code>, <code>/prd-to-plan</code> and on “slice” as the work unit. Moved rule zero early in the entry diagram to match the prose. Added verification pointers for every mechanism claim, a worked grilled-vs-ungrilled spec example, and made the <code>--setting-sources</code> note self-contained.
2.0	2026-08-31	Rewrote as a general adoption playbook: removed session- and repo-specific anecdotes and cost figures in favour of measure-it-yourself guidance. Corrected the Part B interactive workflow order (grill → spec → plan) and each step’s purpose. Rewrote “Where the two layers sit” to describe skill_engineering as shipped — vendoring command, <code>skill_engineering:</code> config key, <code>claude_code</code> -only, per-run cost visibility.

Version	Date	Changes
1.1	2026-08-25	Added five Mermaid diagrams: the entry fork, the gate-verification loop, the four-jobs decision tree, the interactive-vs-headless split, and where the two layers sit today versus after skill_engineering.
1.0	2026-08-25	Initial playbook: existing-codebase adoption, new-project bootstrap, and the standing checklist.